CYLINDER HEAD COVER(S) (Continued)

(1) If a replacement cover is installed, transfer the CCV valve grommet and oil filler cap from the original cover to the replacement cover.

(2) Install cylinder head cover and gasket (Fig. 14). Tighten the mounting bolts to 10 N·m (85 in. lbs.) torque.

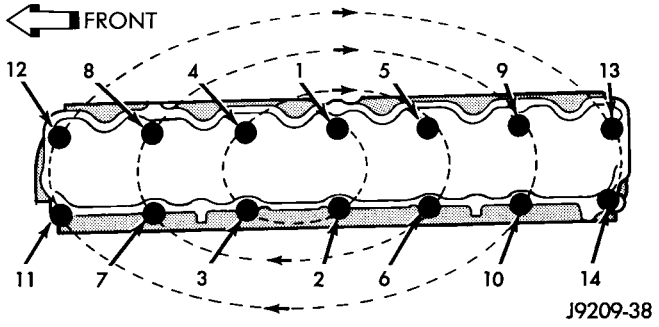

Fig. 14 Cylinder Head Cover Gasket Locator Pins at #8 & #9

(3) Reinstall the plastic wiring loom trough at the rear of cylinder head cover.

(4) Reconnect the left hand side of the wiring harness loom to all components.

(5) Connect the CCV hoses.

(6) Install control cables and bracket on intake manifold and tighten bolts to 8.7 N·m (77 in. lbs.) torque.

(7) Connect control cables to throttle body linkage.

(8) Snap control cables into cylinder head cover clip.

(9) Connect negative cable to battery.

INTAKE/EXHAUST VALVES & SEATS

DESCRIPTION

The valves are made of heat resistant steel and have chrome plated stems to prevent scuffing. All valves use three bead locks to promote valve rotation (Fig. 15).

STANDARD PROCEDURE - VALVE SERVICE

VALVE REFACING

Clean all carbon deposits from the combustion chambers, valve ports, valve stems, valve stem guides and head.

(1) Use a valve refacing machine to reface the intake and exhaust valves to the specified angle.

(2) After refacing, a margin of at least 0.787 mm (0.031 inch) must remain (Fig. 16). If the margin is less than 0.787 mm (0.031 inch), the valve must be replaced.

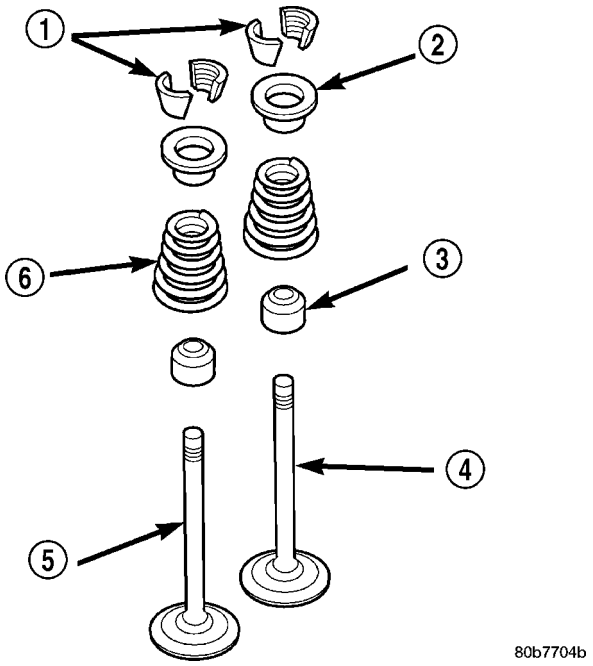

Fig. 15 VALVE AND KEEPER CONFIGURATION 4.0L

1 - VALVE LOCKS (3–BEAD)
2 - RETAINER
3 - VALVE STEM OIL SEAL
4 - INTAKE VALVE
5 - EXHAUST VALVE
6 - VALVE SPRING

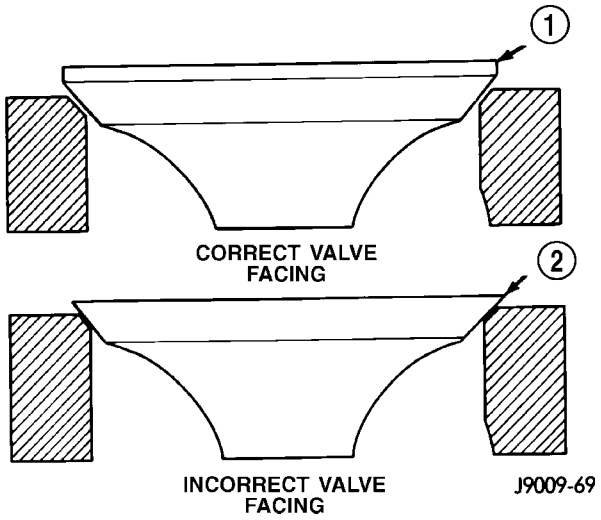

Fig. 16 Valve Facing

1 - VALVE MARGIN
2 - NO MARGIN

VALVE SEAT REFACING

Clean all carbon deposits from the combustion chambers, valve ports, valve stems, valve stem guides and head.

(1) Install a pilot of the correct size in the valve guide bore. Reface the valve seat to the specified